

Smart Portable Real Time Detection for Microplastics in Waterbodies



A MINI PROJECT II REPORT

Submitted by

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In partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

Department of Electronics and Communication Engineering

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

ANNA UNIVERSITY : CHENNAI 600025

April 2026



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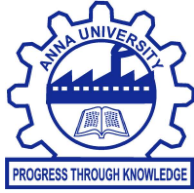
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BONAFIDE CERTIFICATE

Certified that this project report entitled “**Smart portable real time detection for microplastics in waterbodies**” is the bonafide work of **ABARNA C (71812302001), AMRUTHA VARSHINI N (718123302013), KAVITHA S (71812302062)**, who carried out the **20EC280** Mini Project II under my supervision.

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ABSTRACT

Microplastic contamination has become an environmental problem which not only affects water quality but can pose serious health concerns for both humans and aquatic life. Traditional approaches to testing for microplastics tend to be costly, time consuming, and may require special equipment. Therefore, this research project seeks to develop a low-cost portable system for detecting microplastics using optical sensing technology. The light source in this system will be provided by a white LED light, while an Light Dependent Resistor (LDR) will be used to detect changes in light due to the presence of microplastics in the water sample. The suspended particles block or change the path of the light, resulting in changes in the output from the LDR sensor. Other than the optical sensing approach, a turbidity sensor will be included in order to gauge the water turbidity. The last sensor included in this system is an ultrasonic sensor which will help gauge the water height. The information gathered by sensors is processed through an ESP32 microcontroller, which employs threshold logic in detecting the presence of microplastics and calculating their parameters including number, turbidity value, and concentration. All the calculations can be viewed on the LCD screen in real time, which makes the system user-friendly and appropriate for outdoor use. The system can detect the presence of microplastics at sizes from 100 to 300 μm on the basis of the light variations. Experimental findings reveal that it is essential to create appropriate lighting conditions and align the sensors properly in order to increase the detection precision. In conclusion, the developed system provides a simple, inexpensive, and efficient way to monitor microplastics.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iv
	LIST OF FIGURES	vii
	LIST OF ABBREVIATIONS	viii
1	INTRODUCTION	1
2	LITERATURE REVIEW	6
3	PROBLEM IDENTIFICATION AND METHODOLOGY	12
	3.1 PROBLEM IDENTIFICATION	12
	3.2 OBJECTIVES	13
	3.3 METHODOLOGY	13
	3.4 PLAN OF EXPERIMENTS	17
	3.5 THEORETICAL BACKGROUND	18
4	DESIGN AND SIMULATION	19
	4.1 HARDWARE COMPONENTS	19
	4.1.1 WHITE LED	20
	4.1.2 LDR(LIGHT DEPENDENT SENSOR	20
	4.1.3 TURBIDITY SENSOR	21
	4.1.4 ULTRASONIC SENSOR	22
	4.1.5 ESP 32 MICROCONTROLLER	23
	4.1.6 LCD DISPLAY	23
	4.1.7 VOLTAGE REGULATOR	24
	4.1.8 POWER SUPPLY ADAPTER	25
	4.1.9 TRANSPARENT CHAMBER	26

	4.2 DESIGN	27
	4.2.1 STEPS INVOLVED IN DESIGN OF THE PROTOTYPE	27
	4.3 SIMULATION	28
	4.3.1 MACHINE LEARNING	29
	4.4 EXPERIMENTAL STUDY	30
	4.5 CODING	31
5	RESULTS AND DISCUSSION	50
	5.1 TESTING METHODS	50
	5.2 TESTING STANDARDS	50
	5.3 SAMPLE PREPARATION	51
	5.4 HARDWARE SETUP	52
6	CONCLUSION AND FUTURE SCOPE	54
	6.1 CONCLUSION	54
	6.2 FUTURE SCOPE	55
	REFERENCES	
	SDGA GOALS – MAPPING	
	ANNEXURE I - BASE PAPER	
	ANNEXURE II – ACHIEVEMENTS	
	ANNEXURE III - PLAGIARISM REPORT	

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1	Flowchart of proposed methodology	19
4.1	Hardware components	24
4.2	White LED	25
4.3	LDR (Light dependent sensor)	25
4.4	Turbidity sensor	26
4.5	Ultrasonic sensor	27
4.6	ESP 32 Microcontroller	28
4.7	LCD Display	28
4.8	Voltage regulator	29
4.9	Power supply adapter	30
4.10	Transparent chamber	31
4.11	Design of the proposed idea	32
4.12	Machine learning output	33
4.13	Machine learning output	34
4.14	SEM analysis of water sample	35
4.15	SEM analysis of water sample	35
5.1	Hardware setup	38

LIST OF ABBREVIATIONS

FTIR	Fourier Transform Infrared Spectroscopy
ESP32	Espressif 32-bit Microcontroller
LED	Light Emitting Diode
LDR	Light Dependent Resistor
LCD	Liquid Crystal Display
SEM	Scanning Electron Microscope
IoT	Internet of Things
NTU	Nephelometric Turbidity Unit
ADC	Analog To Digital Converter
I2C	Inter-Integrated Circuit
DC	Direct Current
μm	Micrometer
SVM	Support Vector Machine
CNN	Convolutional Neural Network
YOLOv5	You Only Look Once version 5
DeepSORT	Deep Simple Online and Realtime Tracking
RSC	Royal Society of Chemistry
CCWC	Computing and Communication Workshop and Conference

CHAPTER-1

INTRODUCTION

Microplastics are small pieces of plastic, usually less than 5 millimeters in size, which are usually produced by the degradation of larger plastic materials or are specifically made for use in products such as cosmetics, textiles, and industrial products. During the last few decades, the use of plastic has shown a significant increase. Consequently, plastic waste has accumulated in the environment. When these plastics degrade because of environmental factors such as exposure to sun, temperature, and other mechanical forces, microplastics are produced. Microplastics have been found in large quantities in the oceans, rivers, lakes, and even in drinking water supplies.

The occurrence of microplastics in aquatic environments has raised significant environmental concerns. Microplastics have shown adverse effects on the environment. Plastics are not easily degraded like other organic materials. They can persist in the environment for several hundred years. Microplastics are small enough to be consumed by aquatic animals such as fish, planktons, shellfish, and other aquatic organisms. They can accumulate in the food chain and can ultimately enter the human body through the consumption of contaminated water supplies and seafood. Studies have indicated that microplastics can contain harmful chemicals and microorganisms, which can pose health risks to human beings. Scientists have been working hard to investigate the sources, distribution, and impacts of microplastics in the environment. It is, however, very hard to identify and measure the level of microplastics in water samples. It is because they are smaller in size, come in different shapes, and are transparent.

In recent times, with the awareness of environmental pollution, the demand for effective tools to monitor the level of pollution in the environment is on the increase. It is important to monitor the level of pollution in the environment to understand the level of pollution, the sources of pollution, and the ways to reduce the pollution caused by plastics. Governments, environmental agencies, and researchers are working hard to come up with new technologies to reduce the pollution caused by plastics. Recent developments in embedded technology, optical sensors, and machine learning techniques have brought new opportunities to design tools to monitor the level of pollution caused by plastics. These techniques can be used to design tools to monitor the level of pollution caused by plastics efficiently. It is possible to design tools to monitor the level of pollution caused by plastics using both hardware and software techniques.

Due to the growing level of microplastics in water bodies, it is important to develop effective detection and monitoring techniques. Current techniques, although effective, are not applicable for large-scale and field-based detection. Current techniques involve the use of laboratory equipment such as Fourier Transform Infrared Spectroscopy (FTIR), Raman Spectroscopy, and Electron Microscopy. These techniques are very costly and require expertise to operate. Moreover, these techniques involve complex processes and take a long time to complete. Therefore, it is important to develop a system to effectively detect microplastics in water in real time without the need to involve complex techniques. It is important to note that the developed system is able to detect particles with a particular size.

It is important to note that the size of the particles to be detected is between 100 μm to 300 μm . Moreover, the developed system is important to ensure it can operate effectively in different environmental conditions.

Another significant factor that should be considered is the cost-effectiveness of the system. Most of the current detection systems are costly, making it difficult for small research institutions, educational bodies, and developing countries to afford the system. The cost-effectiveness of the system can be achieved by the use of cheap components such as microcontrollers, sensors, and camera modules. All these components can help in reducing the overall cost of the system.

The device should also be small in size and light in weight. This will help in transporting the device to remote areas for use in microplastic monitoring. The use of a portable device will help in reducing the transportation of microplastic particles from the site of collection to the site of analysis. The use of a portable device will help in quick analysis of the microplastic particles, thus helping in quick decision-making on environmental issues.

Apart from the hardware requirements, the system should also have the ability for smart data processing. The use of smart data processing can help in improving the accuracy of the system in microplastic detection. Smart data processing can help in reducing human error in the analysis of microplastic particles. In addition, the system should be user-friendly, meaning that it should not require a lot of technical knowledge to operate. This will be achieved if the system has a simple interface and output results.

Thus, the need to create a low-cost, portable, and intelligent system to detect microplastics arises to overcome the limitations of current methods. This project has been created to meet these requirements, employing optical sensing, embedded systems, and machine learning technologies.

In conclusion, the importance of the detection is related to the protection of the environment. It is important to ensure the quality of the water. By being able to detect the pollution caused by the microplastics, it is possible to take the required steps to control the pollution.

The system that is proposed is the Low-Cost Portable Microplastic Detector with the ability to detect microplastic particles within the 100-300 μm range within the water sample. The system is also able to detect the microplastic particles within the water sample with the help of optical sensing as well as image processing techniques. The system is composed of both hardware as well as software components, which work together to detect the microplastic particles within the water sample. The hardware components used in the system include the ESP32 microcontroller, the camera module, the white light-emitting diode, the laser diode, the photodiode sensor, as well as the flow cell chamber. All the hardware components are arranged in such a way that an optical setup is formed, which is capable of detecting the microplastic particles within the 100-300 μm range.

The flow cell chamber is an important part of the system, as it acts as an environment for the water sample. The water sample is made to flow through the chamber, where the light from the light-emitting diode as well as the laser beam act on the water sample.

The photodiode sensor detects the scattered light due to the particles in the solution. The signal from the photodiode is amplified by the operational amplifier. Then the signal is sent to the ESP32 microcontroller for further processing. This gives information about the particles in the water.

The images are then processed by software algorithms to detect the particles. Various image processing techniques are employed for this purpose. The size, shape, and count of the particles can be found through image processing techniques. Machine learning algorithms are used to classify the detected particles. It is then checked whether the particles are microplastics or not. Thus, the system uses a combination of both techniques for more accurate results. The proposed system is found to be compact, portable, and cost-effective for the detection of microplastics.

The microplastic detection system that has been proposed has several advantages over the traditional microplastic detection methods. The first advantage of the microplastic detection system is that it is cost-effective. The use of ESP32, sensors, and other camera modules in the microplastic detection system makes it cost-effective because these components are not very expensive.

The system is also green technology because it does not require the use of harmful reagents. The system is thus more environmentally friendly compared to other microplastic detection techniques.

CHAPTER-2

LITERATURE SURVEY

M. A. B. Sarker, U. Butt, M. H. Imtiaz, and A. B. Baki (2023), “Automatic Detection of Microplastics in the Aqueous Environment,” *Proceedings of the IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC)*, pp. 1–6.

The automatic microplastic detection system proposed in this research aims to identify microplastics in water samples by utilizing modern image processing and deep learning methods. The proposed microplastic detection system utilizes a camera-based approach to acquire real-time images of water samples containing microplastics. The acquired images are processed by the YOLOv5 object detection algorithm, which is a widely known and highly accurate method for detecting small objects. The YOLOv5 object detection algorithm is trained on labeled datasets of microplastics to improve the detection accuracy of the proposed microplastic detection system. Moreover, the proposed microplastic detection system utilizes the DeepSORT tracking algorithm to track the movement of microplastics in the video frames. The tracking of microplastics helps prevent duplicate counting of microplastics and provides information about the flow and velocity of microplastics. The proposed method significantly reduces labor and increases efficiency compared to conventional methods used in a lab setting, such as microscopy and spectroscopy. The system can detect different shapes and sizes of microplastics in water, making it versatile for different types of water. The use of deep learning algorithms increases the accuracy and flexibility of the system. This research has shown that automated systems can play a significant role in monitoring microplastic pollution in the environment and can contribute to the development of smart and portable systems

P. Murugan, R. Karthikeyan, and S. Kumar (2023), “Recent Advances in Microplastic Detection Technologies,” *RSC Advances*, vol. 13, pp. 21045–21060.

One of the major developments that have been highlighted in the survey is the integration of machine learning algorithms. Algorithms such as Support Vector Machine (SVM), Decision Tree, and Convolutional Neural Networks (CNN) are being used for the automatic classification of microplastics. The survey also highlights the need for the development of portable detection systems. The survey suggests that the integration of sensors and intelligent algorithms could lead to the development of efficient and cost-effective solutions. Overall, this survey highlights the role of new technologies in the detection of microplastics and is a positive step for the development of efficient monitoring systems. This survey will discuss recent developments in technologies for detecting microplastic, with a focus on new technologies that are more accurate and efficient in their results. The survey will discuss the shortcomings of traditional laboratory-based techniques and new techniques such as optical sensing, fluorescence, and machine learning. Optical sensing techniques rely on light-based techniques for detecting microplastic in water. Light scattering and fluorescence are used to detect microplastic particles based on their size and composition. The techniques are non-destructive and allow for real-time monitoring. The survey will discuss how image processing techniques can be used to analyze images and determine features such as size and shape of microplastic particles.

Abimbola, J. Zhou, and L. Chen (2024), “In-Situ Detection of Microplastics in Aquatic Environments: Technologies and Challenges,” *Science of the Total Environment*, vol. 920.

This research is concerned with in-situ detection techniques for microplastics. In-situ detection techniques allow for the identification of microplastics in aquatic environments without the need for further analysis. The research has highlighted the need for in-situ monitoring of microplastics in order to understand the flow of microplastics in natural water bodies. The survey has discussed different optical sensing techniques such as light scattering and fluorescence. These techniques allow for the identification of microplastics by interacting with light. The techniques allow for the constant monitoring of water samples. The use of sensors in these techniques makes it possible for the techniques to be used in the field. The use of in-situ detection techniques has several advantages. In-situ techniques do not require the transportation of water samples.

In most cases, the transportation of water samples from one place to another can cause contamination. The use of in-situ techniques ensures reliable results. Challenges such as environmental noise, water conditions, and the presence of other particles are also addressed in the study. It is suggested that advanced signal processing and machine learning techniques can be used to overcome the challenges. In the research, the potential of portable and automated systems is emphasized in the context of environmental monitoring. It is concluded that in-situ detection techniques can contribute to the efficiency of microplastic monitoring.

M. Campos-López, V. Garibay-Febles, and J. A. Mendoza-Pérez (2025), “Development of a Portable Optical Sensor for Microplastic Detection,” *Applied Sciences*, vol. 15, no. 9.

This survey will discuss the development of a portable optical sensor, which is designed for detecting microplastic particles in water samples. The system will utilize optical sensors, which are capable of detecting microplastic particles in water based on their interaction with light. The system will incorporate a light source and a photodetector, which will work together to measure changes in light intensity due to the presence of particles in the water sample. This system will allow for the detection of microplastic particles in water without having to undergo complex chemical analysis. The survey will show that optical sensors are a viable option for detecting microplastic particles in water, unlike other traditional systems. The system will be tested under different conditions to determine its performance and accuracy in detecting microplastic particles in water. The system will be capable of detecting microplastic particles in water and determining their concentration in the water sample.

The survey also talks about the advantages of portability and ease of use. This compact design enables the system to be used in remote locations, making it suitable for environmental monitoring. The cost of the components is also low, making it accessible to researchers and students. This research supports the development of portable detection systems, and it emphasizes the importance of optical sensing techniques in microplastic detection. This research is a step forward in improving environmental monitoring systems, and such systems will be more efficient in the future.

Z. Huang, X. Zhao, and J. Wang (2022), “Analytical Methods for Microplastics in the Environment: A Review,” *Science of the Total Environment*, vol. 806, pp. 150–163.

The survey is a comprehensive review of the different methods of analysis for microplastics in the environment. The survey discusses traditional methods of analysis such as Fourier Transform Infrared Spectroscopy (FTIR), Raman spectroscopy, and microscopy. The methods are widely used for the analysis of microplastics due to their high accuracy and reliability. However, the methods require special and costly equipment and expertise to carry out the analysis. The survey also discusses the advantages and disadvantages of different methods of analysis. The traditional methods of analysis are accurate for the analysis of microplastics. However, the methods are costly and time-consuming. The methods also require special expertise and equipment for analysis. The survey also discusses some of the challenges faced in the analysis of microplastics. The challenges include contamination of samples, difficulties in identifying small particles of microplastics, and different shapes and sizes of microplastics. In addition to these techniques, the survey also presents the newly developed techniques such as thermal analysis, fluorescence-based techniques, and image processing techniques. These techniques can help in the improvement of the efficiency of the detection process. It is also suggested in the survey that machine learning techniques can be combined with image processing techniques to achieve the goal of the detection process. In the context of the application of the techniques, the importance of developing low-cost and portable techniques is highlighted in the survey. It is suggested that the application of optical sensing techniques combined with computational techniques can help achieve the required balance between cost and efficiency. It can be said that this survey is one of the best references to understand the current techniques.

SUMMARY:

Traditional methods such as FTIR, Raman spectroscopy, and microscopic analysis have shown high accuracy in identifying microplastics and their chemical composition. However, these methods involve high cost, time, and laboratory facilities, making them inappropriate for real-time applications. The recent studies have shown developments in the design of an automated and intelligent system to detect microplastics using image processing and machine learning techniques. Deep learning techniques such as YOLOv5 have been implemented to detect and track microplastics in water, reducing time and manual intervention. These systems have shown real-time monitoring capabilities with accurate results. Optical sensing techniques, including light scattering and fluorescence, have been developed as promising approaches to overcome the challenges in the detection of microplastics. These techniques can be used to perform non-destructive and rapid analysis. Moreover, the incorporation of image processing techniques can be used to improve the accuracy of the sensors. In-situ techniques can be used to eliminate the need for transporting the samples. Portable optical sensors can be developed to improve the efficiency of environmental monitoring. These techniques can be used to perform real-time analysis directly in the water. As indicated in the literature, the application of optical sensors, image processing techniques, and machine learning can be used to address the challenges in the detection of microplastics. These advances can be used to develop efficient and portable techniques for the detection of microplastics, which is important in the protection of the environment.

CHAPTER 3

METHODOLOGY AND WORKING PRINCIPLE

3.1 PROBLEM IDENTIFICATION

- **Lack of Availability of Effective Detection Methods** - The current techniques for detecting microplastics make use of expensive scientific equipment such as spectroscopy and microscopy, making on-the-go or real-time detection not possible.
- **Expensive Existing Methods** - High costs of installing and running existing technologies, such as FTIR and Raman spectroscopy, mean that these methods can only be used in research labs.
- **Lengthy Detection Process** - Traditional methods entail long periods of time before detection can be done, owing to sample collection, processing, and analysis, which may take hours or even days.
- **Dependence on Chemical Staining or Sample Processing** - This entails that the existing technology: Is expensive to implement, Needs skilled technicians and has the potential to change the sample.
- **Lack of Efficient Field Techniques** - Existing methods lack compact and portable tools capable of: Field detection of microplastics, Instantaneous results and Minimal technical knowledge.

3.2 OBJECTIVES:

- **Creation of Light Particle Detection Device** - To create an instrument where the presence of microplastic particles will be detected through variation of light intensity using LED and LDR sensor.
- **Detection with Turbidity Sensor for Better Results** - To combine the use of a turbidity sensor to measure the clarity of water to ensure accurate detection of the particles present.
- **Real-Time Monitoring via ESP32 Microcontroller and LCD** - To analyze the sensors' data using ESP32 microcontroller and display the results on the LCD screen.

3.3 METHODOLOGY

In the system, the concept involves designing an optical detector chamber for detecting microplastics in water samples based on changes in light intensity, as well as sensor fusion. In the design, the test chamber is made using a transparent glass box, while the white LEDs are fitted around the edges of the test chamber to uniformly illuminate the water sample. An LDR is placed at the opposite end of the light source for monitoring the light passing through the test chamber.

Once the device is activated by the user, water is poured into the test chamber. The LED turns on, resulting in continuous light beam penetration through the water sample. Under normal circumstances without the presence of microplastics, the light beam will have a uniform intensity as perceived by the LDR. But when microplastics ranging from 100–300 μm pass through the light beam, there will be fluctuations in light intensity recorded by the LDR. The change in light intensity is converted into voltage/resistance changes and then processed by the ESP32.

The ESP32 board is capable of acquiring signals from LDR in real time and processing them. A trigger mechanism based on a threshold is employed to identify any sudden change in the intensity of light, which is considered indicative of the presence of particles. Particle density can be calculated using the frequency of particle events detected over time. Furthermore, the magnitude and duration of the change in the light intensity signal are used to estimate the size of the particles to classify them in the desired microplastic size range.

A turbidity sensor is embedded in the chamber for more reliable results. The turbidity sensor provides an alternative measure of suspended particles by measuring the cloudiness of the water. Turbidity measurements can be used to validate the LDR measurements.

The ultrasonic sensor is placed at the top of the container and determines the distance from the water surface; hence, the water level and its volume (liters) can be calculated. This variable plays an essential role in calculating the relative concentration of microplastics since the number of particles is normalized based on the volume of the water.

The data collected from all sensors is analyzed and combined using the ESP32 microcontroller, which generates some vital outputs such as:

- Total number of particles
- Turbidity
- Volume of water
- Size range
- Relative concentration

These outputs are then displayed on the LCD display that is connected to the front part of the device (refer to the figure with “Total Particles”). The system is contained in a small enclosure; thus, it becomes portable, inexpensive, and applicable for water quality analysis without using advanced camera-based imaging technologies.

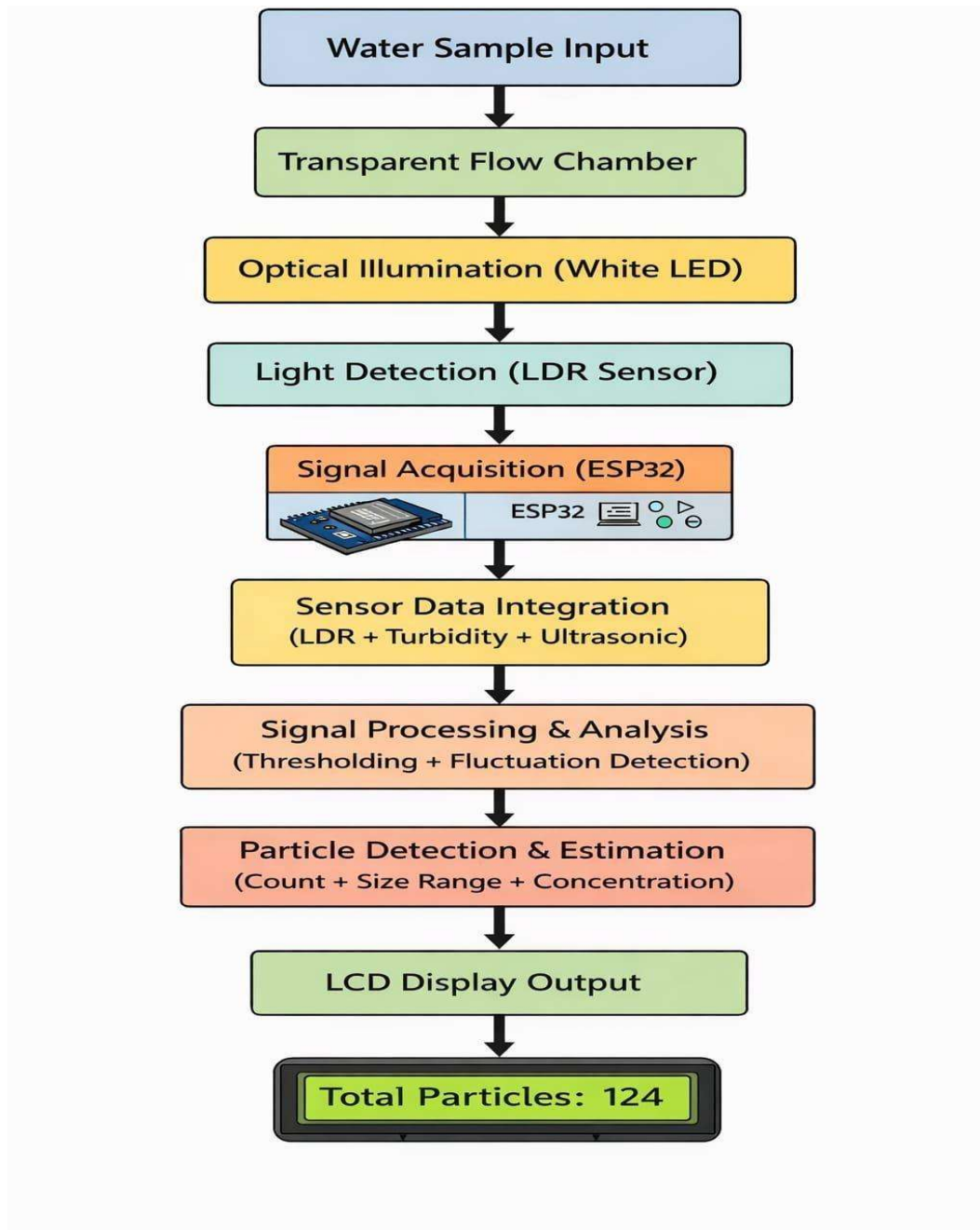


Fig 3.1 Flowchart of proposed methodology

The figure 3.1 illustrates a step-by-step process for real-time detection of microplastic particles in water using optical sensing and multiple sensors. It integrates data from LDR, turbidity, and ultrasonic sensors, which are processed by an ESP32 for accurate particle analysis. Finally, the system estimates particle count, size, and concentration, displaying the results on an LCD output.

3.4 PLAN OF EXPERIMENTS

- **System Setup & Calibration** - Construct the entire system (LED, LDR, turbidity sensor, ultrasonic sensor, ESP32, LCD). Calibration of the LDR in terms of light intensity in: Clean water (baseline), Water containing known particles and Set threshold values in ESP32 to measure light variations.
- **Baseline Experiment With Clear Water** Fill clean/fresh water in the chamber. Measure: LDR signal value, Turbidity reading and Water level (ultrasonic sensor).
- **Experiment With Controlled Particles Added** - Introduce some controlled amount of small particles (fine plastic dust or contamination). Observe: Variation in LDR value (light scattering and blocking due to the particles),
- Increased turbidity reading and Confirm if the system is able to detect particles successfully.
- **Vary With Particle Concentration** - Introduce different amounts of particles in water. Measure: Number of detected particles, Turbidity reading and Check variations based on concentration.
- **Effect of Changing Water Volume with Constant Number of Particles** - Vary water levels in chamber using ultrasonic sensor. Measure effect on: Relative concentration of particles, Confirm that calculation of concentration depends on volume.

- **Repeating for Multiple Trials** - Repeat the experiment several times under identical conditions. Check: Reproducibility of results (number of particles), Stability of sensor readings and Adjusting the system parameters if needed.
- **Output Checking** - Output check of LCD screen: Particle number, Turbidity reading and Water level measurement.

3.5 THEORETICAL BACKGROUND

The microplastic detection system under discussion involves a number of basic concepts, such as optical sensing, light scattering, turbidity measurement, ultrasonic sensing, and signal processing. Light propagates along a straight line while passing through pure water, while its propagation becomes scattered, absorbed, and partially blocked by microplastic particles of sizes from 100 to 300 micrometers (μm), thus causing changes in its intensity. For this particular device, the white LED will serve as the light source, and the LDR as the light detector. The resistance of the LDR changes inversely proportional to the intensity of the light; therefore, with less light being detected because of particle interference, the resistance will become higher, translating into voltage signals.

In order to enhance the accuracy of particle detection, a turbidity sensor is utilized, measuring the turbidity of water, which correlates with the concentration of the particles. Water level is measured via an ultrasonic sensor that works on the time-of-flight principle. Information collected via sensors is further processed by the ESP32 microcontroller based on the thresholding principle and fluctuation analysis of the signal.

CHAPTER 4

DESIGN AND SIMULATION

4.1 HARDWARE COMPONENTS

The design segment illustrates a low-cost portable microplastic detection system using optical sensing and embedded processing. It detects microplastics in water by analyzing light scattering and turbidity variations. The system uses an ESP32 microcontroller to process sensor data and provide real-time output and alerts.

The Design of the hardware includes the hardware description as the basic one. Therefore ,the hardware description is followed by,

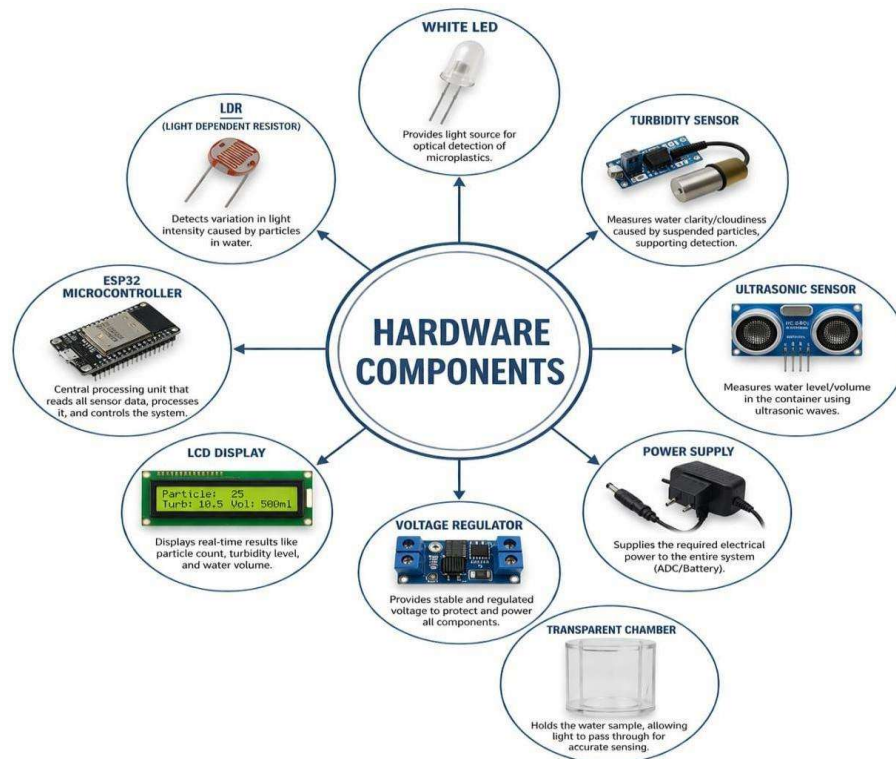


Fig 4.1 Hardware components

Figure 4.1 depicts the hardware components used in the proposed solution.

4.1.1 WHITE LED



Fig 4.2 White LED

Figure 4.2 acts as the major light source in the design. It provides a steady and bright light beam which then shines through the sample of water present in the translucent container. The core mechanism of operation of the design depends on the interaction between light and particles. As such, the use of an LED becomes quite important for optical sensing. The presence of particles means that the light is either scattered, absorbed, or obstructed. All these changes have to be taken into account during the detection process. The use of the LED has several advantages including low energy consumption, high lifespan, compact nature, and uniform illumination. Aligning the LED correctly relative to the LDR is vital for ensuring accurate detection of changes in light intensities. Therefore, the white LED plays a major role in facilitating optical detection of particles in the sample of water.

4.1.2 LDR (LIGHT DEPENDENT RESISTOR)

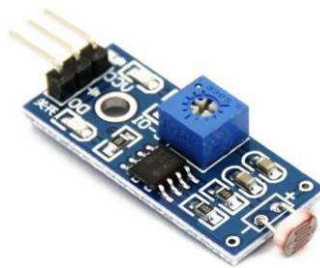


Fig 4.3 LDR (Light dependent resistor)

Figure 4.3 is one of the sensors involved in detecting light intensity variations within the sample once the light gets passed through it. The function of the LDR sensor is based on the premise that the resistance value will reduce when the light intensity becomes high but increases if the light intensity becomes low. The presence of microplastic within the water sample causes scattering or obstruction of the light beam from the LED, causing the LDR to produce varying levels of resistance due to varying intensities of light falling on it. This variation in resistance will then be measured in volts by the ESP32. LDR sensor is inexpensive and efficient since it measures changes in light intensity rather than measuring absolute values of light intensity. LDR stands for Light Dependent Resistor.

4.1.3 TURBIDITY SENSOR



Fig 4.4 Turbidity Sensor

Figure 4.4 is a turbidity sensor which determines the level of cloudiness or haziness in the water due to the presence of particulates. The device functions using the concept of light scattering where light emitted from an internal light source is shone into the water, and the scattered light is determined by the sensor. The higher the turbidity, the higher the number of particulates in the water. In the current project, the turbidity sensor plays the role of being an auxiliary parameter that confirms the presence of microplastics found in the water using the LDR sensor. The device does not distinguish between microplastics and other particulates in the water, it determines the overall quality of the water.

4.1.4 ULTRASONIC SENSOR



Fig 4.5 Ultrasonic Sensor

Figure 4.5 depicts the ultrasonic sensor which is employed for measuring the water level or volume in the tank. It functions using the time-of-flight method, whereby the sensor sends out an ultrasonic wave that is subsequently reflected by the surface of water. The elapsed time before receiving the ultrasonic wave is measured for determining the distance between the water surface and the sensor. This distance will further help determine the water level or water volume. As far as the current project is concerned, the measurement of water volume is necessary for calculating the concentration of microplastics since the number of particles depends on the water volume. The advantage of employing this sensor is that it is non-contact, which means that there is no interaction of the sensor with water, hence its durability. Furthermore, this sensor is easily interfaced with ESP32 and gives accurate readings.

4.1.5 ESP 32 MICROCONTROLLER

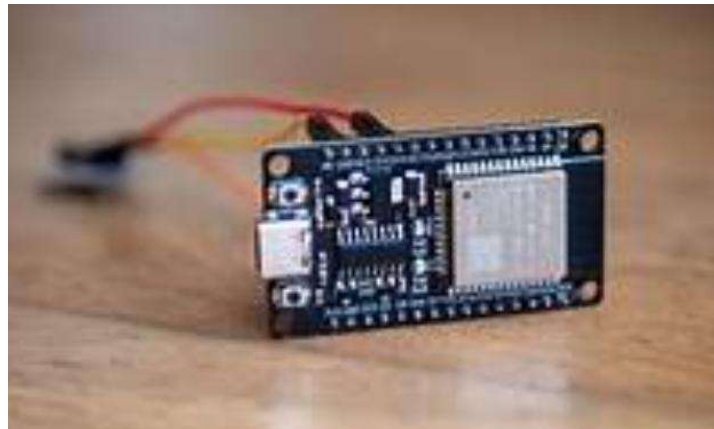


Fig 4.6 ESP 32 Microcontroller

ESP32 Microcontroller: Figure 4.6 is the heart of the entire project. It gathers data from all types of sensors like LDR, turbidity, and ultrasonic sensors. The collected data is further analyzed through the ESP32 microcontroller by using threshold logic and signal processing algorithms to understand variations in light intensity and turbidity levels. After analyzing the data, ESP32 calculates particle count, size of the particle, and relative concentration. ESP32 is selected because it is fast processing, multi- input output, and has in-built Wi-Fi and Bluetooth technology. Future scope can be achieved by adding remote controlling features to the project. This device coordinates all components of the system in real time. The ESP32 also displays the result on the LCD screen.

4.1.6 LCD DISPLAY

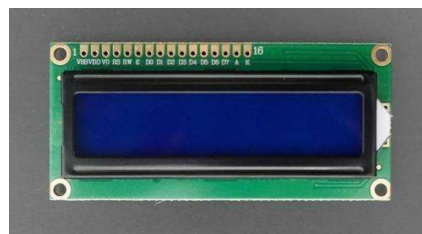


Fig 4.7 LCD Display

The LCD screen is used as an interface that will be utilized for the display of the result obtained from the system. It shows essential variables like particle count, turbidity, and water volume in real-time mode. The LCD screen is connected to the ESP32 that provides processed information that will be displayed on the screen. The use of the LCD screen has been done based on the simplicity, low power requirement, and easy handling nature of the LCD screen.

Figure 4.7 will help the user to have an understanding of the output obtained without requiring other external hardware or software. The real-time feature of the LCD screen makes the proposed system more convenient for practical purposes. The use of the LCD screen increases the usability of the designed system.

4.1.7 VOLTAGE REGULATOR



Fig 4.8 Voltage regulator

Voltage regulator is one of the most important parts which guarantees constant and accurate voltage distribution throughout all system components. Components like ESP32 and various sensors have certain requirements to their operational voltage.

Fluctuations in voltage levels may affect reading results and even cause damages to the components. Figure 4.8 allows maintaining the same level of output voltage despite any input voltage changes.

It also provides protection against overvoltages and any electrical noise. In our project, voltage regulator is important because it will allow maintaining balance, especially when several sensors will work at once. Voltage regulator significantly increases the efficiency of the whole system and its durability. Otherwise, the precision of the results obtained by sensors might suffer.

4.1.8 POWER SUPPLY ADAPTER



Fig 4.9 Power supply adapter

The power supply adapter supplies the needed energy in the form of electricity to power the whole system. This can either be in the form of a battery or an external DC power source based on the design.

The power supply should have the adequate voltage and current levels to run the various parts of the circuit including the ESP32 board, sensors, and display. Power stability is critical in ensuring proper functioning of the circuit since any power interruptions may influence sensor readings and system malfunctioning. Battery-powered power supplies are preferred for portable designs because of their easy portability. Figure 4.9 powers the various parts of the circuit and ensures uninterrupted running of the device.

4.1.9 TRANSPARENT CHAMBER



Fig 4.10 Transparent chamber

The transparent chamber acts as the holding point for the water sample to be tested. It is constructed in such a way that light emitted from the LED can easily penetrate through the water and illuminate the LDR without any interference. The transparency aspect is key because it ensures effective penetration of the light and allows accurate measurement of light intensity changes due to microplastics. Figure 4.10 also serves as the site where the sensing area for turbidity and ultrasonic sensing occurs. It is usually made from transparent materials like plastic or glass to prevent light absorption. The proper design of the chamber ensures uniform dispersion of particles inside the water sample. This makes detection more accurate and reliable while ensuring constant conditions for the experiment.

4.2 DESIGN

The suggested system aims at detecting the presence of microplastics in water using optical detection methods. The use of light scattering, turbidity sensing, and embedded computing forms an important part of the proposed system. The system uses sensors along with a microcontroller for performing real-time analysis of water for microplastics.

Working of the Micoplastic Detection Prototype

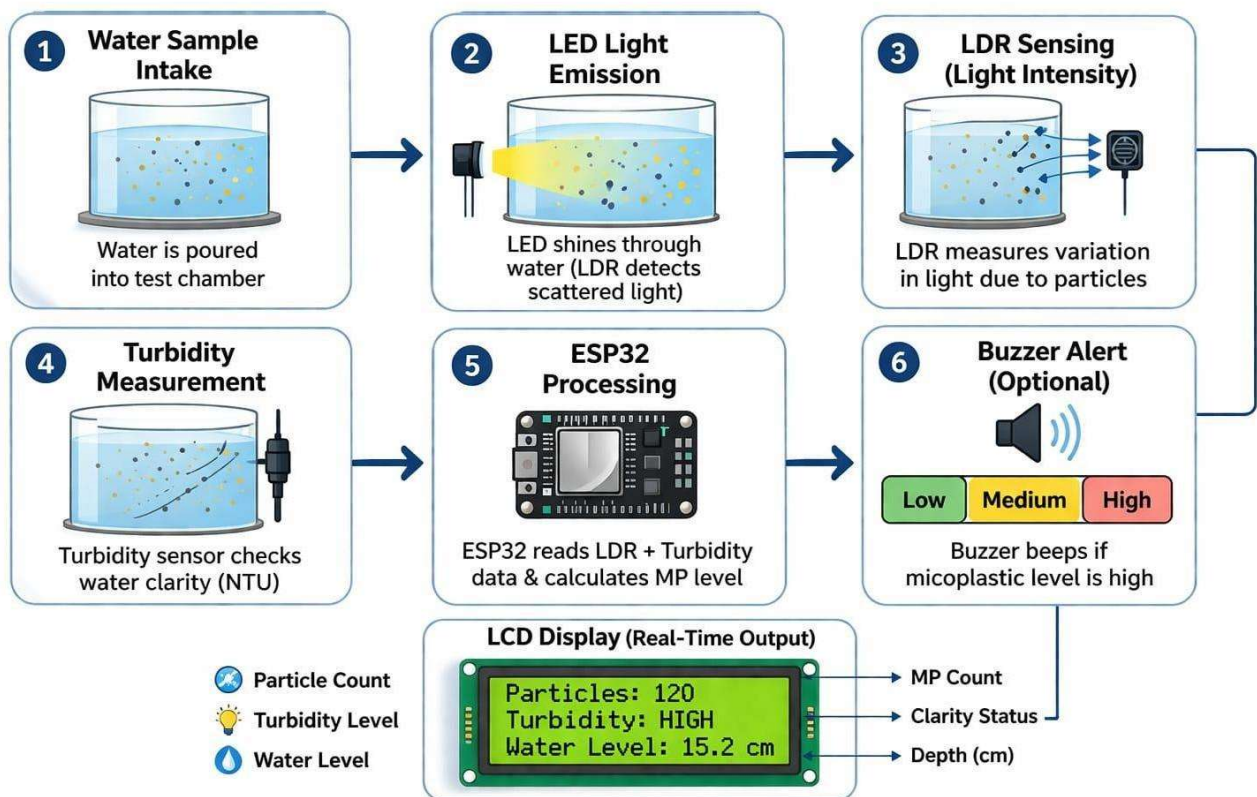


FIG 4.11 Design of the proposed idea

Figure 4.11 depicts the working process of the microplastic detection prototype using optical sensing and embedded processing.

4.2.1 STEPS INVOLVED IN DESIGN OF THE PROTOTYPE

- **Water Sample Collection** - Water sample is collected in the test chamber that contains microplastics (between 100 to 300 μm).
- **Water Turbidity Calculation** - The turbidity sensor detects the clarity of water in NTU units.
- **ESP32 Signal Processing** - ESP32 processes the signal received from the LDR and turbidity sensor and finds the microplastic concentration.
- **Display and Alarm Module** - Data related to microplastic particles, turbidity, and water level is shown on LCD screen.

4.3 SIMULATION

4.3.1 MACHINE LEARNING

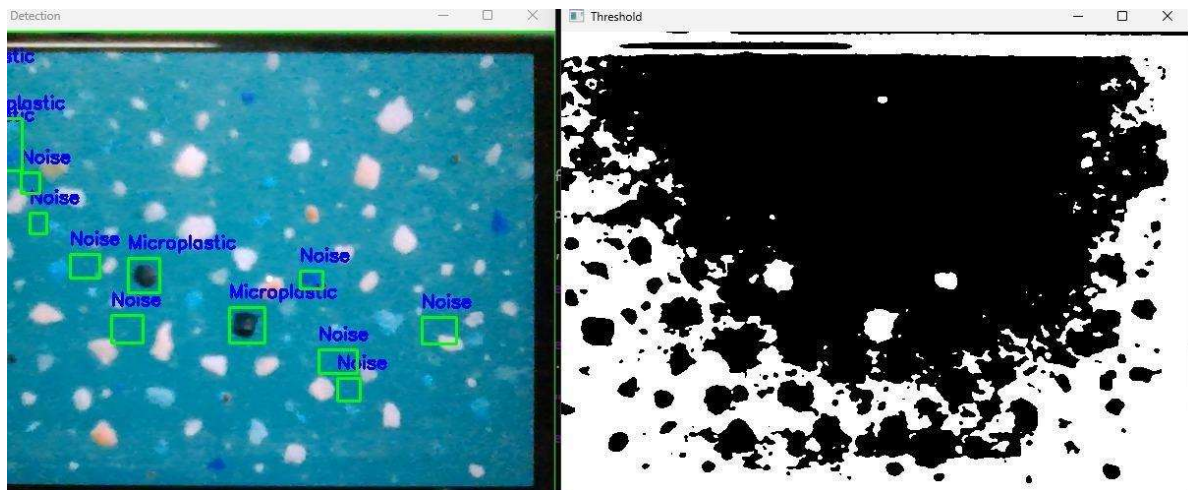


Fig 4.12 Machine learning output

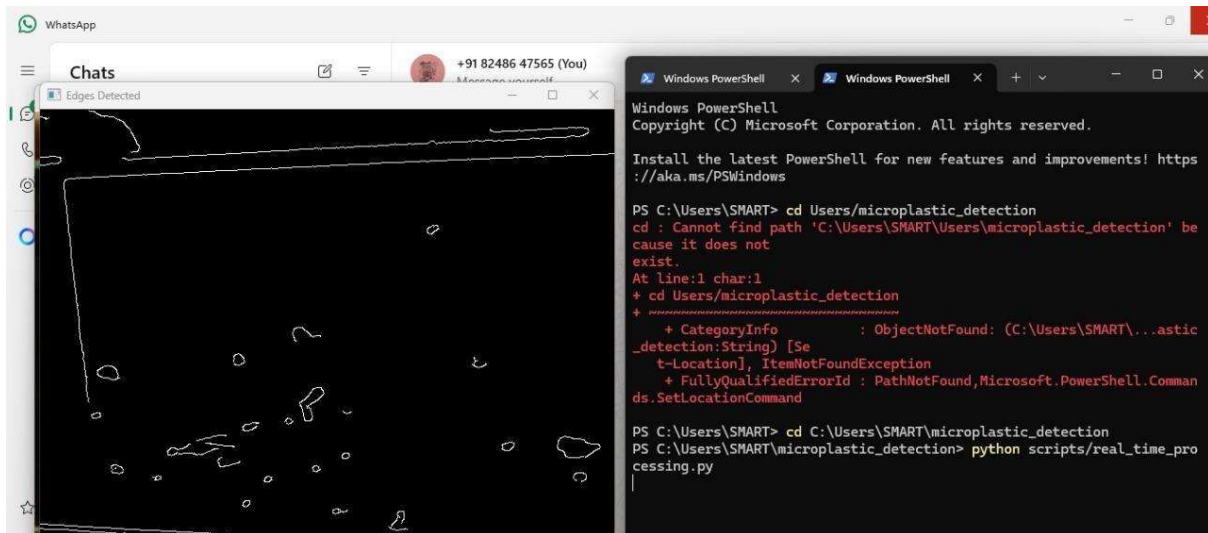


Fig 4.13 Machine learning output using edge detection

Figure 4.12 and Figure 4.13 depicts the machine learning output of the microplastic detection.

The system shows efficiency in detection of the microplastics using the optical sensor in terms of varying the light intensity. Light emitted from the white LED is passed across the liquid samples where it encounters some variations when particles block its path. LDR receives the varying light intensities and processes the data using ESP32 to detect any particle presence.

Analog inputs are converted to digital values and further filtered using thresholding techniques, and only meaningful light variations are considered to ensure accurate detection. Additionally, turbidity levels and water height are considered to increase reliability.

From the above parameters, features that include varying light intensities, turbidity, and water volumes can be extracted to determine particles' presence and distribution. Based on the parameters, the system classifies contaminated water into different levels such as low, moderate, and high.

In the end, output is presented on the LCD screen in real-time showing the estimated number of particles, turbidity levels, and height of water. The system is efficient in detecting particles within the 100-300 μm range approximately.

4.4 EXPERIMENTAL STUDY

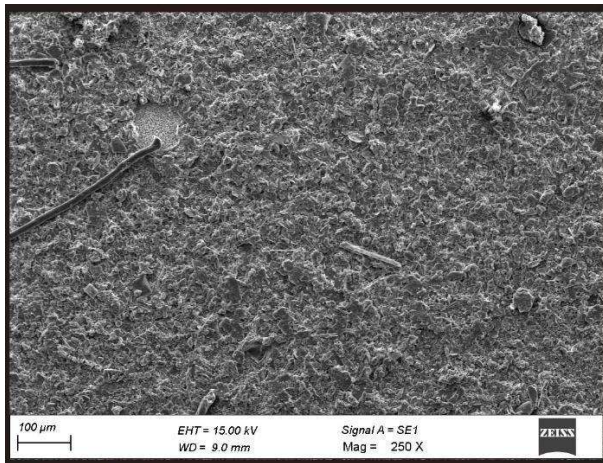


Fig.4.14

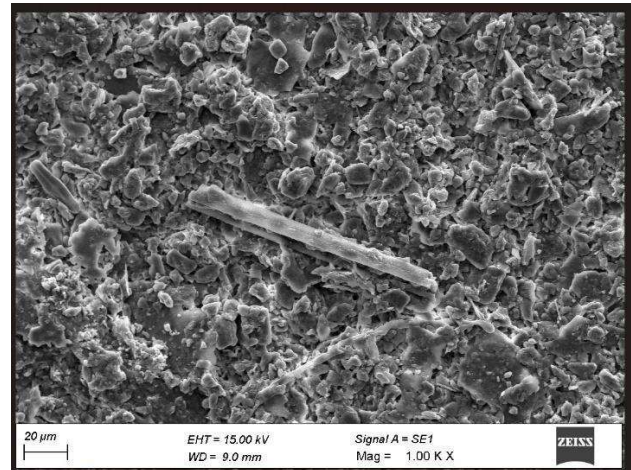


Fig.4.15

Figure 4.14 and Figure 4.15 depicts output obtained from the Scanning electron microscope (SEM) analysis of a water sample.

The presence of heterogeneous particulate material within the tested water samples can be observed under different magnifications through the images taken using the Scanning Electron Microscope (SEM) (250 $\times$, 1.00 k $\times$, and 4.00 k $\times$). The 250 $\times$ magnification reveals that the surface is filled with irregularly distributed particles, demonstrating a high amount of suspension particles.

Under 1.00 k $\times$ magnification, the specific shapes of particles can be seen, including angular fragments and rod-like particles. It should be stated that the elongated shape of the particles indicates fibrous materials that are made of synthetic textiles.

The 4.00 k $\times$ image shows the rough texture of the particles on the surface with irregular edge structures, which are typical for plastic fragments but not natural minerals.

Thus, the SEM test proved the presence of microfibers and fragmented particles, demonstrating contamination.

Consequently, it is necessary to mention that the test confirmed the presence of microplastic-like and other particulate materials in the water sample.

4.5 CODING

```
from machine import Pin, ADC, I2C
import time
import math
from lcd_api import LcdApi
from i2c_lcd import I2cLcd
```

```
# Pin Definitions
```

```
TRIG_PIN = 5
```

```
ECHO_PIN = 18
```

```
TURBIDITY_PIN = 34
```

```
LDR_ANALOG_PIN = 35
```

```
LDR_DIGITAL_PIN = 27
```

```
# LCD Configuration
```

```
I2C_ADDR = 0x26
```

```
I2C_NUM_ROWS = 2
```

```
I2C_NUM_COLS = 16
```

```
# Initialize pins
```

```
trig = Pin(TRIG_PIN, Pin.OUT)
```

```
echo = Pin(ECHO_PIN, Pin.IN)
turbidity_adc = ADC(Pin(TURBIDITY_PIN))
ldr_adc = ADC(Pin(LDR_ANALOG_PIN))
ldr_digital = Pin(LDR_DIGITAL_PIN, Pin.IN)

# Configure ADC
turbidity_adc.atten(ADC.ATTN_11DB)
ldr_adc.atten(ADC.ATTN_11DB)

# Initialize I2C and LCD
i2c = I2C(0, scl=Pin(22), sda=Pin(21), freq=400000)

# Verify LCD is detected
print("Scanning I2C bus...")
devices = i2c.scan()
print(f'I2C devices found: {[hex(device) for device in devices]}')
if 0x26 in devices:
    print("LCD found at address 0x26")
else:
    print("LCD not found at 0x26! Available addresses:", [hex(device) for device in
devices])
# Initialize LCD
lcd = I2cLcd(i2c, I2C_ADDR, I2C_NUM_ROWS, I2C_NUM_COLS)

# Water volume tracking
water_volume_liters = 0
water_height_cm = 0
```



```
def measure_water_volume():
    """Measure water volume in liters using ultrasonic sensor"""
    global water_volume_liters, water_height_cm

    # Send ultrasonic pulse
    trig.value(0)
    time.sleep_us(2)
    trig.value(1)
    time.sleep_us(10)
    trig.value(0)

    # Measure echo time with timeout
    timeout = 30000
    start_time = time.ticks_us()

    while echo.value() == 0:
        if time.ticks_diff(time.ticks_us(), start_time) > timeout:
            return water_volume_liters
    pulse_start = time.ticks_us()

    while echo.value() == 1:
        if time.ticks_diff(time.ticks_us(), pulse_start) > timeout:
            return water_volume_liters
    pulse_end = time.ticks_us()
```

```
# Calculate distance and water level
pulse_duration = time.ticks_diff(pulse_end, pulse_start)
distance = pulse_duration * SOUND_SPEED / 2
```

```
# Calculate water height
water_height_cm = CONTAINER_HEIGHT - distance
if water_height_cm < 0:
    water_height_cm = 0
if water_height_cm > CONTAINER_HEIGHT:
    water_height_cm = CONTAINER_HEIGHT
```

```
# Calculate volume in liters
volume_cm3 = CONTAINER_WIDTH * CONTAINER_LENGTH * water_height_cm
water_volume_liters = volume_cm3 / 1000.0
return water_volume_liters
```

```
def read_turbidity():
    """Read turbidity sensor - returns NTU value (0-1000)"""
    total = 0
    for i in range(SAMPLE_COUNT):
        total += turbidity_adc.read()
        time.sleep_ms(10)
    avg_reading = total / SAMPLE_COUNT
    # Convert ADC reading to NTU
    # Higher ADC = clearer water = lower NTU
    turbidity = (1 - (avg_reading / 4095.0)) * 1000
    return turbidity
```



```

def read_ldr_intensity():
    """Read LDR analog value - returns light intensity percentage (0-100%)"""
    total = 0
    for i in range(SAMPLE_COUNT):
        total += ldr_adc.read()
        time.sleep_ms(10)

    avg_reading = total / SAMPLE_COUNT
    # Higher reading = more light = clearer water
    light_intensity = (avg_reading / 4095.0) * 100
    return light_intensity

def calculate_particle_count(turbidity, light_intensity, water_volume):
    """Calculate realistic particle count based on sensor readings (max 80)"""
    # Base particle count from turbidity (0-1000 NTU range)
    # Adjusted to max 80 particles total
    # Clean water: 0-50 NTU → 0-15 particles
    # Slightly turbid: 50-200 NTU → 15-35 particles
    # Turbid: 200-500 NTU → 35-60 particles
    # Very turbid: 500-1000 NTU → 60-80 particles

    if turbidity < 50:
        # Clean water - very few particles
        base_particles = turbidity * 0.3 # 0-15 particles
    elif turbidity < 200:
        # Slightly turbid - increasing particles

```

```

    base_particles = 15 + (turbidity - 50) * 0.133 # 15-35 particles
elif turbidity < 500:
    # Turbid - more particles
    base_particles = 35 + (turbidity - 200) * 0.083 # 35-60 particles
else:
    # Very turbid - maximum particles (up to 80)
    base_particles = 60 + (turbidity - 500) * 0.04 # 60-80 particles

# Adjust based on light intensity (confirmation factor)
# Higher light = clearer water = fewer particles
if light_intensity > 70:
    adjustment = 0.85 # Reduce particles for clear water
elif light_intensity > 40:
    adjustment = 1.0 # Normal for medium light
else:
    adjustment = 1.15 # Increase particles for turbid water

particle_count = int(base_particles * adjustment)
# Scale by water volume (particles per liter)
particle_count = int(particle_count * water_volume)
# Ensure particle count stays within 1-80 range
particle_count = max(1, min(MAX_PARTICLES, particle_count))
return particle_count

def calculate_avg_size(particle_count, turbidity):
    """Calculate realistic average particle size based on particle count"""

```

```
# More particles = smaller particles
# Clean water = larger particles (more visible)
```

```
if particle_count < 20:
    # Few particles - larger size
    size = 180 - int(particle_count * 2)
elif particle_count < 40:
    # Moderate particles - medium size
    size = 140 - int((particle_count - 20) * 1.5)
elif particle_count < 60:
    # Many particles - smaller size
    size = 100 - int((particle_count - 40) * 1.2)
else:
    # Very many particles - very small size
    size = 70 - int((particle_count - 60) * 1)
```

```
# Adjust based on turbidity
```

```
if turbidity > 500:
    size = int(size * 0.7)
elif turbidity > 300:
    size = int(size * 0.85)
```

```
# Ensure realistic range
size = max(30, min(250, size))
return size
```

```
def calculate_size_range(avg_size):
    """Calculate realistic size range based on average size"""
    size_min = int(avg_size * 0.6)
    size_max = int(avg_size * 1.7)

    # Ensure min is at least 10
    size_min = max(10, size_min)
    size_max = min(450, size_max)
    return size_min, size_max
```

```
def calculate_scattering_level(particle_count):
    """Determine scattering level based on particle count
```

REVERSED LOGIC:

High Scattering = Many particles (55-80)

Medium Scattering = Moderate particles (25-55)

Low Scattering = Few particles (0-25)

"""

```
if particle_count >= 55:
```

```
    return "High"    # High scattering = many particles
```

```
elif particle_count >= 25:
```

```
    return "Medium"  # Medium scattering = moderate particles
```

```
else:
```

```
    return "Low"     # Low scattering = few particles
```

```
def calculate_concentration(particle_count, water_volume):
```

```
    """Calculate particles per mL"""
```

```

if water_volume > 0:
    concentration = particle_count / (water_volume * 1000)
else:
    concentration = 0
return concentration

```

```

def calculate_microplastic_params(water_volume, turbidity, light_intensity):
    """Calculate microplastic parameters based on sensor readings"""

    # CORRECTED LOGIC:
    # HIGH light intensity = CLEAR water = FEW particles = LOW scattering
    # LOW light intensity = TURBID water = MANY particles = HIGH scattering

    print(f"\n--- Sensor Readings ---")
    print(f"Light Intensity: {light_intensity:.1f}%")
    print(f"Turbidity: {turbidity:.1f} NTU")
    print(f"Water Volume: {water_volume:.2f} L")
    # Calculate particle count based on real sensor values
    particle_count = calculate_particle_count(turbidity, light_intensity, water_volume)
    # Determine water clarity based on readings
    if light_intensity > 70:
        water_clarity = "CLEAR"
        print(f"Water: CLEAR → Few particles ( {particle_count} ) → Low Scattering")
    elif light_intensity > 40:
        water_clarity = "SLIGHTLY TURBID"
        print(f"Water: SLIGHTLY TURBID → Moderate particles ( {particle_count} ) →
Medium Scattering")

```

else:

 water_clarity = "TURBID"

 print(f"Water: TURBID → Many particles ({particle_count}) → High Scattering")

Calculate average size based on particle count

avg_size = calculate_avg_size(particle_count, turbidity)

Calculate size range

size_min, size_max = calculate_size_range(avg_size)

Calculate scattering level (REVERSED)

scattering_level = calculate_scattering_level(particle_count)

Calculate concentration (particles per mL)

concentration = calculate_concentration(particle_count, water_volume)

print(f"Particle Count: {particle_count} (Max: {MAX_PARTICLES})")

print(f"Average Size: {avg_size} um")

print(f"Size Range: {size_min}-{size_max} um")

print(f"Scattering Level: {scattering_level}")

return {

 'particle_count': particle_count,

 'avg_size': avg_size,

 'size_min': size_min,

 'size_max': size_max,

 'concentration': concentration,

 'scattering_level': scattering_level,

```
'water_volume': water_volume,  
'turbidity': turbidity,  
'light_intensity': light_intensity,  
'water_clarity': water_clarity  
}
```

```
def display_serial_results(results):  
    """Display detailed results on serial monitor"""  
    print("\n" + "=" * 40)  
    print("MICROPLASTIC ANALYSIS")  
    print("=" * 40)  
    print(f"Total Particles: {results['particle_count']}")  
    print(f"Average Size: {results['avg_size']} um")  
    print(f"Size Range: {results['size_min']}-{results['size_max']} um")  
    print(f"Concentration: {results['concentration']:.2f} particles/mL")  
    print(f"Scattering Level: {results['scattering_level']}")  
    print(f"\nSensor Data:")  
    print(f"Water Volume: {results['water_volume']:.2f} L")  
    print(f"Light Intensity: {results['light_intensity']:.1f}%")  
    print(f"Turbidity: {results['turbidity']:.1f} NTU")  
    print("=" * 40 + "\n")
```

```
def display_lcd_results(results):  
    """Display results on LCD exactly in the format shown"""  
  
    # Screen 1: MICROPLASTIC ANALYSIS title  
    lcd.clear()
```

```
lcd.move_to(0, 0)
lcd.putstr("MICROPLASTIC")
lcd.move_to(0, 1)
lcd.putstr("ANALYSIS")
time.sleep(2.5)
```

Screen 2: Total Particles

```
lcd.clear()
lcd.move_to(0, 0)
lcd.putstr("Total Particles:")
lcd.move_to(0, 1)
lcd.putstr(str(results['particle_count']))
time.sleep(2.5)
```

Screen 3: Average Size

```
lcd.clear()
lcd.move_to(0, 0)
lcd.putstr("Average Size:")
lcd.move_to(0, 1)
lcd.putstr(f"{results['avg_size']} um")
time.sleep(2.5)
```

Screen 4: Size Range

```
lcd.clear()
lcd.move_to(0, 0)
lcd.putstr("Size Range:")
```



```
lcd.move_to(0, 1)
lcd.putstr(f" {results['size_min']} - {results['size_max']} um")
time.sleep(2.5)
```

Screen 5: Concentration

```
lcd.clear()
lcd.move_to(0, 0)
lcd.putstr("Concentration:")
lcd.move_to(0, 1)
lcd.putstr(f" {results['concentration']:.2f} particles/mL")
time.sleep(2.5)
```

Screen 6: Scattering Level

```
lcd.clear()
lcd.move_to(0, 0)
lcd.putstr("Scattering Level:")
lcd.move_to(0, 1)
lcd.putstr(results['scattering_level'])
time.sleep(2.5)
```

Screen 7: Water Volume

```
lcd.clear()
lcd.move_to(0, 0)
lcd.putstr("Water Volume:")
lcd.move_to(0, 1)
lcd.putstr(f" {results['water_volume']:.2f} L")
time.sleep(2.5)
```

```
# Screen 8: Water Clarity
```

```
lcd.clear()
```

```
lcd.move_to(0, 0)
```

```
lcd.putstr("Water Clarity:")
```

```
lcd.move_to(0, 1)
```

```
lcd.putstr(results['water_clarity'])
```

```
time.sleep(2.5)
```

```
# Summary Screen
```

```
lcd.clear()
```

```
lcd.move_to(0, 0)
```

```
lcd.putstr(f"P: {results['particle_count']} S: {results['avg_size']}um")
```

```
lcd.move_to(0, 1)
```

```
lcd.putstr(f"V: {results['water_volume']:.2f}L {results['scattering_level']}")
```

```
time.sleep(3)
```

```
def display_countdown(seconds_left, water_volume):
```

```
    """Display countdown timer on LCD with water volume"""
```

```
    lcd.clear()
```

```
    lcd.move_to(0, 0)
```

```
    lcd.putstr("Next Analysis:")
```

```
    lcd.move_to(0, 1)
```

```
    lcd.putstr(f"{seconds_left}s | V: {water_volume:.2f}L")
```

```
def perform_analysis():
```

```
    """Perform complete microplastic analysis"""
```

```
# Read sensors
```

```
water_volume = measure_water_volume()
```

```
turbidity = read_turbidity()
```

```
light_intensity = read_ldr_intensity()
```

```
# Calculate parameters
```

```
results = calculate_microplastic_params(water_volume, turbidity, light_intensity)
```

```
# Display results
```

```
display_serial_results(results)
```

```
display_lcd_results(results)
```

```
return results
```

```
def calibrate_sensors():
```

```
    """Initial calibration sequence"""
```

```
    print("Calibrating sensors...")
```

```
    lcd.clear()
```

```
    lcd.move_to(0, 0)
```

```
    lcd.putstr("Calibrating...")
```

```
    for i in range(5):
```

```
        time.sleep_ms(100)
```

```
        lcd.move_to(0, 1)
```

```
        lcd.putstr(f"Step {i+1}/5")
```

```
    print(f"\nCalibration complete!")
```

```
    print(f"Container:
```

```

{CONTAINER_WIDTH}x{CONTAINER_LENGTH}x{CONTAINER_HEIGHT} cm")
    print(f"Maximum Volume: {MAX_VOLUME} Liters")
    print(f"Maximum Particles: {MAX_PARTICLES}")
    print(f"\n--- Scattering Level Logic ---")
    print(f"Particles 0-25 → Low Scattering")
    print(f"Particles 25-55 → Medium Scattering")
    print(f"Particles 55-80 → High Scattering")
    print(f"\n--- Water Quality Logic ---")
    print(f"High Light (>70%) = CLEAR WATER = Few Particles = Low Scattering")
    print(f"Low Light (<40%) = TURBID WATER = Many Particles = High Scattering")

```

```
def main():
```

```

    """Main program loop - 15 second interval"""
    print("\n" + "=" * 50)
    print("MICROPLASTIC ANALYSIS SYSTEM")
    print("ESP32 MicroPython - Real Sensor Values")
    print(f"Container:
{CONTAINER_WIDTH}x{CONTAINER_LENGTH}x{CONTAINER_HEIGHT} cm")
    print(f"Maximum Volume: {MAX_VOLUME} Liters")
    print(f"Maximum Particles: {MAX_PARTICLES}")
    print("=" * 50)

    # Calibrate sensors
    calibrate_sensors()

    # Measure initial water volume
    print("\nMeasuring initial water volume...")

```

```
initial_volume = measure_water_volume()
initial_light = read_ldr_intensity()
initial_turbidity = read_turbidity()
print(f'Current water volume: {initial_volume:.2f} Liters')
print(f'Light Intensity: {initial_light:.1f}%')
print(f'Turbidity: {initial_turbidity:.1f} NTU")
# LCD initialization
lcd.clear()
lcd.move_to(0, 0)
lcd.putstr("Microplastic")
lcd.move_to(0, 1)
lcd.putstr("Analyzer Ready")
time.sleep(2)
analysis_count = 0
# Main loop
while True:
    try:
        analysis_count += 1
        current_volume = measure_water_volume()
        current_light = read_ldr_intensity()
        print(f'\n{'='*50}')
        print(f'Analysis #{analysis_count} - {time.localtime()}')
        print(f'Water Volume: {current_volume:.2f} Liters')
        print(f'Light Intensity: {current_light:.1f}%')
        print(f'{'='*50}')
        # Perform analysis
        results = perform_analysis()
```

if results:

 # Display completion message

 lcd.clear()

 lcd.move_to(0, 0)

 lcd.putstr("Analysis Done!")

 lcd.move_to(0, 1)

 lcd.putstr("Next in 15 sec")

 time.sleep(1.5)

 # Countdown for 15 seconds

 print(f"\nNext analysis in 15 seconds...")

 for remaining in range(15, 0, -1):

 # Update countdown every second with water volume

 current_vol = measure_water_volume()

 display_countdown(remaining, current_vol)

 time.sleep(1)

except KeyboardInterrupt:

 print("\n\nAnalysis stopped by user")

 lcd.clear()

 lcd.move_to(0, 0)

 lcd.putstr("System Stopped")

 lcd.move_to(0, 1)

 lcd.putstr("Thank you!")

 break

except Exception as e:

```
print(f"Error: {e}")  
lcd.clear()  
lcd.move_to(0, 0)  
lcd.putstr("Error occurred")  
lcd.move_to(0, 1)  
lcd.putstr("Restarting...")  
time.sleep(5)
```

Run the program

```
if __name__ == "__main__":  
    main()
```

CHAPTER 5

RESULTS AND DISCUSSION

5.1 TESTING METHODS

- **Clear Water Testing** - Test using clear water to determine the baseline values of LDR and turbidity (no particles present).
- **Particle Detection Testing** - Place small particles in the system and see whether the LDR can detect light changes and accurately count the particles.
- **Turbidity Validation Test** - Compare turbidity values against the particles present to confirm accuracy.
- **Water Level Testing** - Utilize the ultrasonic sensor to test for various levels of water volume measurement.
- **Repeatability Testing** - Repeating the experiments ensures stability and consistency of results.
- **Output Validation** - Verify the outputs displayed on the LCD monitor.

5.2 TESTING STANDARDS

- **Calibration Standard** - Sensors (LDR, turbidity, ultrasonic) are calibrated by use of reference conditions such as clean water and certain levels. To ensure that sensors are accurate.
- **Reproducibility Standard** - Same experiment is done under the same conditions repeatedly. To ensure that the system gives similar and reliable outputs.
- **Accuracy Test** - Outputs from the system (particles, turbidity, water level) are compared to the expected or known readings. To measure accuracy of the outputs.

- **Sensitivity Test** - System performance with different concentration levels of particles (low to high). To determine sensitivity of the system.
- **Turbidity Reference Standard** - Turbidity readings are compared to reference clarity standards (clear, moderately turbid, and highly turbid). To ensure valid interpretation of data.
- **Functional Test** - Components (LDR, LED, ESP32, and display unit) are tested for functionality individually and collectively. To ascertain the functional ability of all components.

5.3 SAMPLE PREPARATION

The preparation of water samples for microplastic particle detection using the proposed detection system was done through very simple and straightforward processes that imitate actual field scenarios. First, clean or distilled water was used as a base sample for which readings for light intensity, turbidity, and water volume were noted.

Plastic particles are manually scraped or crushed to produce microplastic particles from various plastics like water bottles and other plastic containers. These particles are then manually sieved or picked out to obtain particles whose diameters fall in the range of 100-300 microns. These particles are then added to the distilled water sample at specific concentrations, i.e., low, medium, and high concentrations.

These samples are then mixed thoroughly to distribute particles evenly across the water sample while keeping the water volume constant for each experiment for easy comparison of results. The mixture is left to settle before conducting tests to eliminate any air bubbles.

Finally, the water sample is added to the transparent container used to detect microplastic particles.

5.4 HARDWARE SETUP

The discussion section includes the final prototype developed and its description:

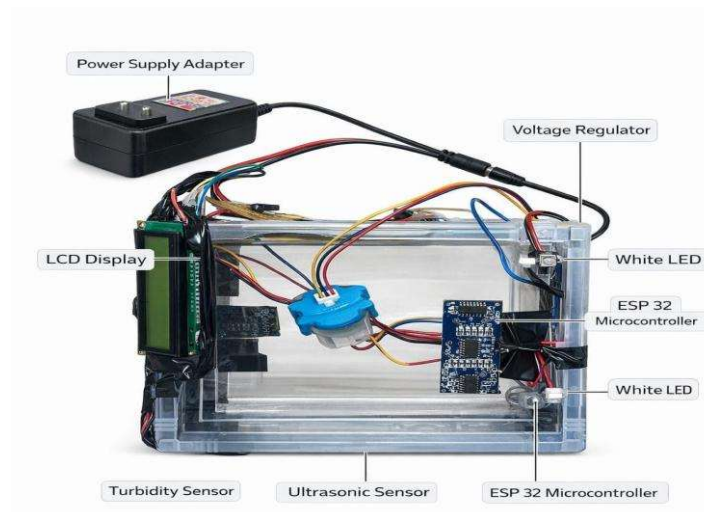


Fig 5.1 Hardware setup

Figure 5.1 is the Prototype developed which is compact and integrated smart monitoring system comprising of an ESP32 microcontroller. System development was done in such a way that it can be viewed from outside and monitored at any time.

Power supply to the whole system is provided using an external adapter. Voltage regulator ensures stable voltage distribution among the different parts of the system. As a result, this improves the efficiency of the system and prevents any damage which might have been caused by voltage fluctuations.

The turbidity sensor used in the system helps in measuring the amount of clarity or dirtiness of the liquid inside the chamber. This makes the system ideal for use in the process of water quality measurement. In addition to the turbidity sensor, the ultrasonic sensor is used in determining the liquid level in order to ensure continuous monitoring and avoid overflow or dry run situations.

The presence of the LDR and the white LED shows that illumination is used in some aspects to control sensing process. Lastly, the LCD serves as a screen where real-time sensor measurements are displayed. ESP32 microcontroller serves as the main processor in the system.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

In the present scenario, microplastic pollution is recognized as an important issue in the environment, aquatic life, and health. The rise in the concentration of microplastic material in water bodies calls for the immediate need for the development of an efficient system for the detection of microplastics. The traditional methods used for the detection of microplastics, such as FTIR and Raman spectroscopy, are highly accurate but not cost-effective. This project aims to develop an efficient system for the detection of microplastics.

In the present project, the system for the detection of microplastics is developed with the help of optical sensing techniques and image processing methods. The hardware part consists of various components such as the white light-emitting diode, laser diode, photodiode sensor, ESP32 microcontroller, camera module, etc. The software part is an important part of the system, as the detected data is processed with the help of an image processing algorithm, followed by the implementation of machine learning techniques. It help in the classification of the detected microplastic particles, resulting in the accuracy of the system. The machine learning algorithm helps in the classification of the detected particles. The system is highly efficient in the real-time detection of microplastic particles in the range of 100-300 μm .

One of the advantages of the proposed system is its cost-effectiveness and portability. Unlike the traditional methods, the proposed system can operate in field environments such as rivers, lakes, and wastewater treatment plants. Moreover, the proposed system eliminates the need for complex chemical preparation, making it user-friendly and applicable to different fields.

The results obtained using the proposed system can be useful for environmental research activities. Moreover, the proposed system shows the potential for using embedded systems with artificial intelligence to address real-world problems.

In conclusion, the proposed system for detecting microplastics is effective and useful. Although the proposed system is limited to the size of the particles it can detect and the optical conditions required, it sets the stage for future innovations. With the incorporation of advanced technologies, this system can contribute to the improvement of water quality and the environment.

6.2 FUTURE SCOPE

The proposed low-cost portable microplastic detection system is a good foundation for further development and improvement of technologies in the field of environmental monitoring systems. Although the proposed microplastic detection system is good for detecting microplastics of sizes ranging from 100 to 300 μm , there is a significant scope for improvement and development of such systems in the future.

One of the major areas of improvement for such systems is the detection of smaller microplastics of sizes less than 100 μm . The microplastics of smaller sizes are harmful because they have the capability of entering biological systems easily. The proposed microplastic detection system can be improved by equipping it with better camera capabilities and better image processing techniques for the detection of smaller microplastics of sizes less than 100 μm .

Another major area of improvement for such systems is the use of better machine learning and deep learning techniques for microplastic detection. The proposed microplastic detection system is based on lightweight machine learning algorithms such as Decision Trees and Support Vector Machines.

The microplastic detection system can be improved by equipping it with better machine and deep learning capabilities such as convolutional neural networks and object detection techniques such as YOLO.

The system can further be enhanced through the integration of Internet of Things (IoT) technology, which enables the device to be connected to cloud platforms, thus allowing data collected from different locations to be stored, analyzed, and monitored remotely. This will enable authorities to monitor trends in pollution over time.

Furthermore, enhancements in the design of the device will make the system more robust and efficient for use in monitoring water pollution. Miniaturization of components, compact design, and efficient power management will enable the device to operate effectively in the field over long periods of time. In addition, solar power can be integrated with the system, which will enable its use in areas where there is no power.

Finally, with many devices deployed in rivers, lakes, and wastewater treatment facilities, this system will provide valuable information for studies in water pollution, which will enhance its potential for use in monitoring microplastic pollution in water bodies worldwide, with continuous enhancements to its design, functionality, and robustness to become a smart system in the near future.

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SDGA GOALS



The developed system of microplastic detection plays an important role in ensuring the achievement of Sustainable Development Goal number 6 (Clean Water and Sanitation), as the development allows carrying out water quality analysis using affordable technologies that will be able to detect pollution caused by the presence of microplastics. Moreover, the project is relevant in terms of Sustainable Development Goal 12 (Responsible Consumption and Production) because it creates opportunities to make people aware of plastic pollution and promote proper waste management procedures.

ANNEXURE 1

BASE PAPER

Automatic Detection of Microplastics in the Aqueous Environment

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Abstract— Microplastics (<5 mm) have become a global concern due to their growing threat to the marine and freshwater environment. There is a lack of technologies for the rapid and accurate identification and quantification of microplastics in the aqueous environment. This paper presents a deep-learning-based methodology for real-time detection, tracking, and counting of microplastics in freshwater environments through real-time object detection. A prototype was developed to detect microplastics of 1 mm to 5 mm in size and different shapes (e.g., spherical) and colors (e.g., red, green, blue). The microplastics detection model employed the small YOLOv5 architecture as we focused on low-power applications. In-situ image collection was performed using a Logitech C270 camera, and the microplastics were manually annotated on those images before being applied for model training. For real-time object tracking, we used Simple Online and Real-time Tracking with a Deep Association Metric (DeepSORT), an extended version of the Simple Online and Real-time Tracking (SORT) algorithm. Our developed system can work up to 34 cm/sec of water velocity and successfully detect, track, count, and calculate the velocity of microplastic of size 5mm.

Keywords—Deep learning, DeepSORT, Microplastics, YOLOv5.

I. INTRODUCTION

Microplastics (MP's), defined as plastic debris having sizes < 5 mm and > 1 μm [1], are a global ecological problem of increasing scientific concern and are recognized as one of the major environmental concerns by the United Nations [2]. Without significant action, there would be more plastics than fish in the ocean, by weight, by 2050 – as noted by World Economic Forum [3]. Microplastics act as vectors of contaminants in water bodies and are alarming in the human body through biological processes. The microplastic contaminant can enter trophic food webs [4] and pose a significant threat to human health and ecosystems. A recent study [5] found that 1000 rivers globally transport 0.8–2.7 million metric tons of plastic litter each year to the ocean. A significant proportion (about 70-80%) of marine/lake plastic debris/microplastics is assumed to be derived from riverine systems [6]. Nevertheless, most of the literature on microplastics has focused on marine organisms and habitats, with fewer studies [7], [8] conducted in riverine systems. Since riverine microplastics research is still in its infancy, many uncertainties remain [9]. Given the high degree of complexity in riverine systems (e.g., changing climate, land cover, flow

regimes, geographical factors), riverine microplastics detection techniques, dynamics, and biological interactions remain relatively under-researched. Further, it is necessary to develop appropriate management strategies and feedback platforms to mitigate riverine microplastic contamination.

Researchers use different types of methods to detect microplastics; Microscopy, Fourier transform infrared spectroscopy, Raman spectroscopy, thermal analysis, etc. [10]. Fluorescence microscopy [11], light microscopy [12], stereomicroscope [13], and scanning electron microscopy (SEM) [14] are often employed techniques for the detection of microplastics [10]. Thermal analysis can analyze the thermal stability or thermal degradation of polymers to determine information about their composition [15]. Majewski et al.[16] used thermal analysis to determine PP and PE microplastics. However, all of these methods are laboratory-based and require the water samples from the water body to be carried to the controlled lab environment. Computer vision has recently become increasingly popular for detecting microplastics [17]. It was found that detailed, enlarged photographs of surfaces can provide structural information that can be used to detect microplastics [18]. The motivation is from numerous camera-based modern applications, including autonomous vehicles, security and surveillance, and medical imaging. In the current domain, Machine learning and Deep learning-based approach, as well as classical computer vision techniques, have been employed to detect microplastics. Although computer vision is used to detect microplastic by researchers, real-time detection is not implemented for fresh water.

The contribution of this study is the presentation of a camera-based system for real-time detection, counting, and getting the travel path of microplastics in fresh water. In this project, we focused on riverine freshwater microplastic detection. To detect microplastics, a YOLOv5-based object detection model was developed. As we are counting real-time moving objects, we have implemented DeepSORT to track the object. We have also calculated the velocity and travel path of microplastics that can help to determine the microplastics coverage areas at a certain time.

This paper is organized as follows: Section II describes the methodology. The experimental setup is described in Section III. In section IV, image acquisition and training methods are

described. The result and discussion are in section V, and the conclusion is in section VI.

II. METHODOLOGY

After capturing, the frame is passed to the YOLOv5 object detection model. If any microplastic is detected, the frame is passed to the DeepSORT to track the microplastic. Then DeepSORT assigns a unique id to each microplastic. We placed two imaginary vertical lines on the frame. One line (line 2) is at 50 % of the image's width, and another one (line 1) is 100px before that line. We start counting when the object passes the first line (line 1). When it passes the second line (line 2), we calculate the velocity of the microplastic, which we calculate from the length and time it takes to pass the area inside those imaginary lines. We count the microplastic, and the count and velocity are shown on the screen and saved into a text file. Visualization helps to analyze the real-time scenario, and saved files will be useful later. If the YOLOv5 model does not recognize any microplastics, DeepSORT cannot assign any id to the microplastic. In that case, it simply shows the input images and doesn't save any count. Fig. 1. shows the flowchart of the developed system.

A. Detection and Model Development

We have implemented YOLOv5 in this project, an updated version of the well-known object identification and image segmentation model YOLO (You Only Look Once) [19]. The performance of the YOLO model was further enhanced, and additional capabilities were added in 2021 with the release of YOLOv5 by Ultralytics [19]. Researchers have implemented YOLOv5 in lots of applications, including real-time marine object detection [20], traffic detection [21], and safety helmet detection [22]. And in this project, we found promising results using this model.

For tracking the object, we have implemented Deep Association Metric (DeepSORT) [23]. DeepSORT is a deep learning-based multiple-object tracking algorithm. Estimating the trajectory of multiple objects simultaneously over time in a series of video frames is known as multiple object tracking (MOT). One such quite straightforward tracking-by-detection technique is Simple Online And Real-time Tracking (SORT) [24]. For simplicity and speed, it ignores the appearance features in the association matrix and instead uses a mix of the Kalman Filter and the Hungarian algorithm to handle motion prediction and data association, respectively. This increases the number of identity switches. DeepSORT incorporates item appearance information in an association matrix to make up for this performance loss [25]. In particular, DeepSORT applies a convolutional neural network (CNN) to overcome the problems of SORT. DeepSORT assigns a unique id for each object detected in the frame.

B. Velocity of Microplastic

The velocity of the microplastic in the flowing water was calculated after detection. The velocity of microplastic is calculated in cm/second. The velocity varies depending on the

microplastics' size, weight, and shape. The following equation is used:

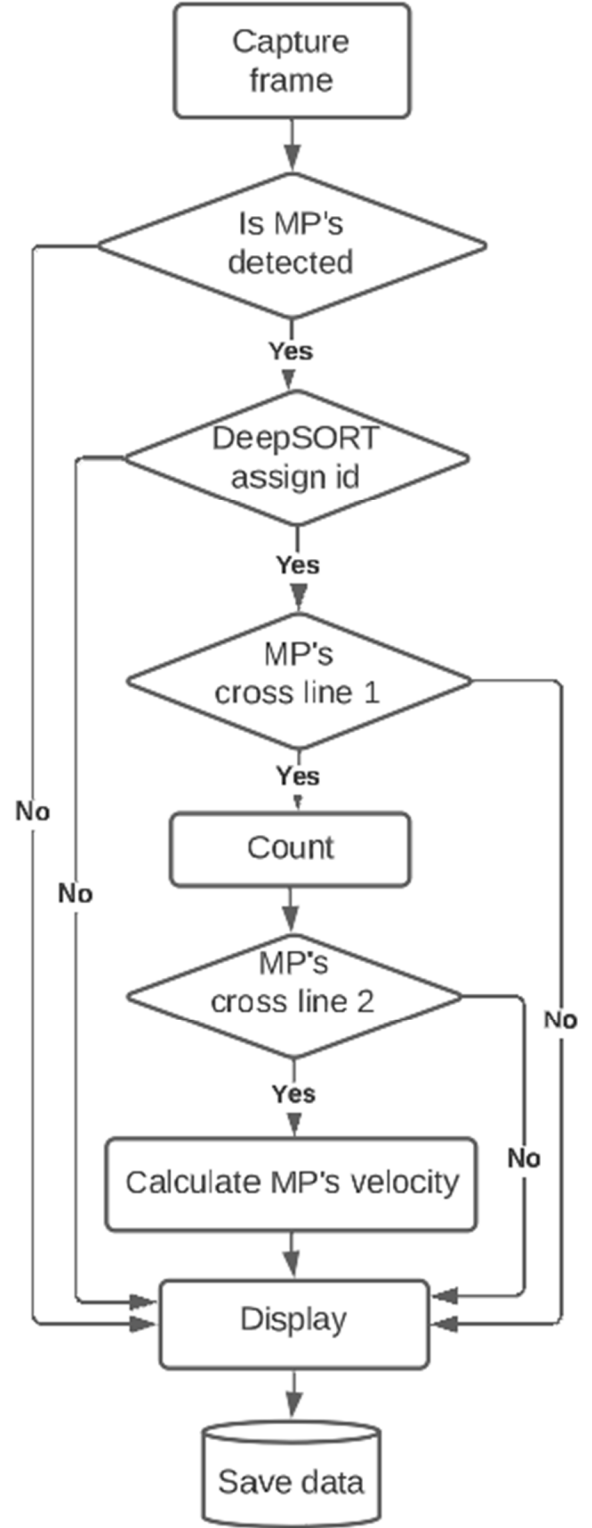


Fig. 1. Flowchart of the developed methodology

$$v = \frac{W_{cc} \times \Delta xp}{W_p \times \Delta t} \quad (1)$$

Where v is the average velocity of microplastic W_{cc} is the camera converge width in cm, Δxp is the displacement of microplastic in pixels, W_p is the image's width in pixels, Δt is changed in time seconds.

III. EXPERIMENTAL SETUP

The experiments were performed in the 12-meter long and 0.45-meter wide, recirculating open channel flume in the water resources lab at Clarkson University. A transparent flume sidewall was used with an adjusted slope of $S_0 = 0.005$. The longitudinal, transverse, and vertical dimensions of the flume were denoted by x , y , and z directions, respectively. An observation area of 2.00×0.46 m was selected at a distance of 4.99 m away from the start of the flume to ensure a fully developed turbulent flow (Fig. 3). The flow rate was recorded by using pressure gauges connected to a computer. The flow depth was measured and adjusted using a point gauge and tailgate at the flume end. An Acoustic Doppler Velocimeter (ADV) was used to measure water velocity. Three different

categories of microplastics were used: low-density polystyrene (PS) spheres (4.9 mm), high-density microplastic spheres made of Cellulose Acetate (CA) (2.7 mm) and Acrylic Polymethyl – Methacrylate (PMMA). were released at the centerline of the flume where the ADV readings were taken. Ten microplastics of each type were released simultaneously with the help of the funnel attached to the point gauge, where the bottom part was always kept 0.02 m below the water surface to avoid surface turbulence. After release, the microplastics were collected in a fine mesh (0.5 mm opening size) positioned at the end of the flume. Fig. 2, 3, and 4 show the experimental setup flume plan view, experimental setup, and image of the experimental setup.

The movement of the microplastics was tracked with the Logitech C270 Camera. It is a fixed-focused camera that can take 720 pixels of video at 30fps [26]. The Logitech C270 webcam was selected as the system's vision sensor because of its low price, compact design, and USB interface. Logitech C270 camera is not waterproof, so we added waterproof superglue to the holes and joints to make it watertight. We submerged the camera in the water to run the experiment and capture the required images.

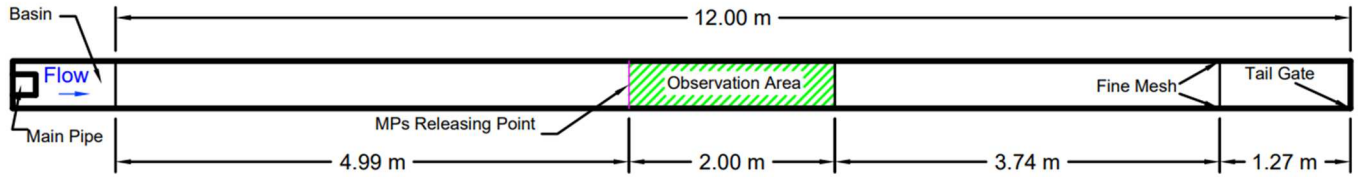


Fig. 2. Flume plan view (flume measurements).

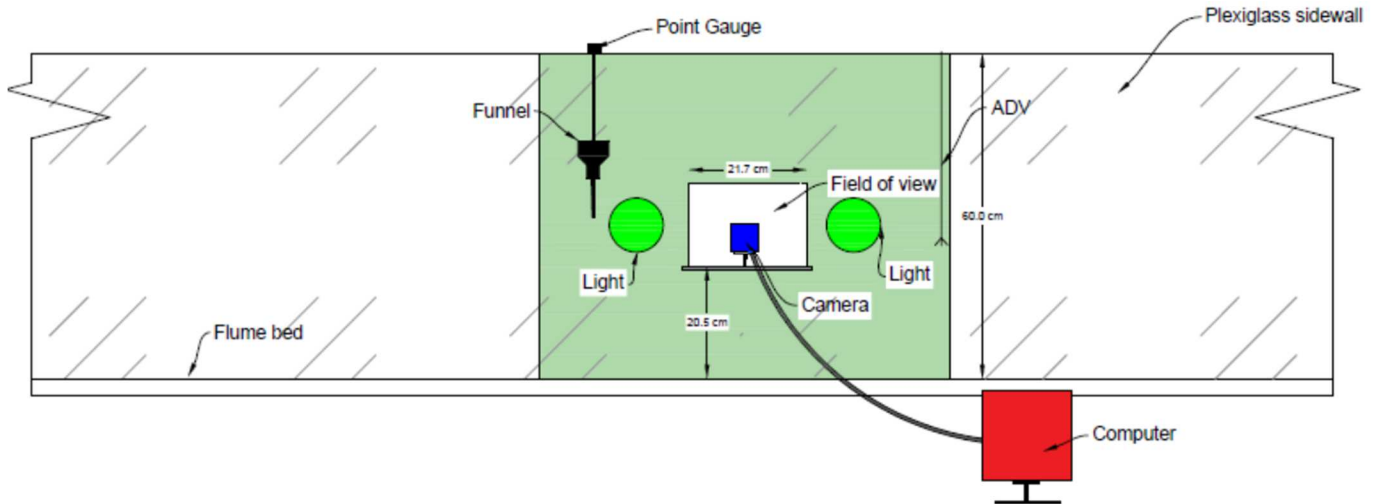


Fig. 3. Schematic Representation of Experimental setup for this study

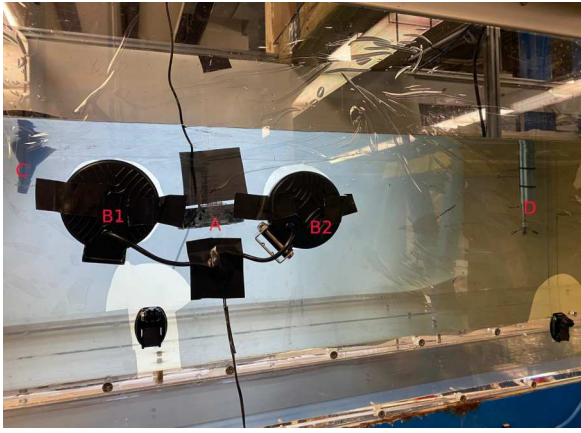


Fig. 4. Experimental setup A: Camera, B1, and B2: LED light, C: funnel D: Acoustic Doppler Velocimeter (ADV).

IV. IMAGE ACQUISITION AND TRAINING

Using the experimental setup shown in Fig. 2 - 4, we collected 2000 images of microplastic of size (1 mm and 5 mm) and colors (cyan, pink). The image resolution was 640x480 pixels. After collecting all the images, all microplastics in the images were annotated using the open-source software Labellmg [27]. We have separated the images training, test, and validation (80%,10%,10%). After separating the images, we trained using YOLOv5. The total annotated image was 2000, the batch size was 16, the learning rate was 0.01, and the epoch was 100. In testing, our results, described in the result section, look promising.

V. RESULT AND DISCUSSION

Fig. 5. shows the microplastics detection, count, travel path, and velocity. As microplastics are moving, we implemented DeepSORT beside YOLOv5 to track the microplastics. We found that the DeepSORT assigned id changes frequently for smaller objects like microplastics. So, we placed two imaginary lines in the frame. When the microplastics passed line 1, we started counting, and we calculated velocity when microplastics passed line 2. The total count value is shown in the top-left corner of Fig. 5. We ran the inference with the collected video. Selecting a good camera that works well underwater for real-time detection was challenging. We plan to detect real-time microplastic in flowing water where microplastic will flow with the tide. Microplastics are so small, and we used a tradeoff between power and resolution to get a good image. If we take a large image, we will get good features of objects, but at the same time, training and processing time will be longer. We ran the test in the lab environment and had considerable results. Our target is to run the study on any free water source and verify the technology in a real environment.

We placed 5 mm spherical cyan color microplastic on the flowing water at different velocities, as shown in Table 1. Our system can detect up to 35 cm/sec; faster than that, accuracy reduces until 50 cm/sec when the system cannot detect any 5 mm size microplastics. With 1 mm microplastic, the system can accurately detect water velocity up to 20 cm/sec and fails to detect over 30 cm/sec.

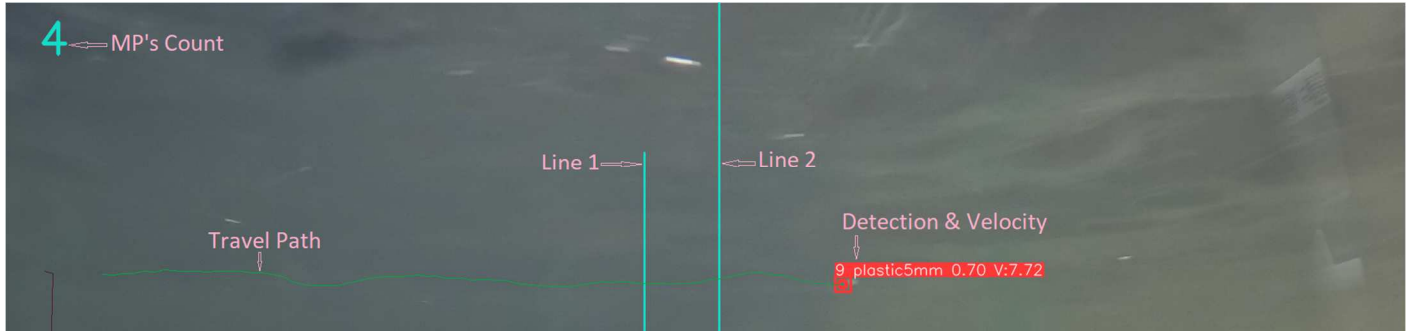


Fig 5: Microplastic detection count, path travel path, velocity calculation

As we detect microplastic in flowing water, it is necessary to understand how the system works in different water velocities.

We took 5 mm spherical cyan color microplastic and placed it on the flowing water at different velocities shown in Table 1. Our system can detect up to 35 cm/sec; faster than that, accuracy reduces until 50 cm/sec when the system cannot detect any 5 mm size microplastics. With 1 mm microplastic, the system can accurately detect water velocity up to 20 cm/sec and fails to detect over 30 cm/sec.

TABLE I
MICROPLASTIC COUNTING ACCURACY ON CERTAIN WATER VELOCITIES

Sl	Water Velocity (cm/sec)	Accuracy (5mm)	Accuracy (1mm)
1	1.07	100%	100%
2	20	100%	100%
3	34	100%	0%
4	50	0%	0%

Although the detection and tracking of microplastics are accomplished in this paper, the suggested method still has certain drawbacks, and there is room for improvement. We are working on different sizes and shapes of microplastic and different water velocities for further improvement.

VI. CONCLUSION

This paper has investigated vision-based microplastic detection and tracking using in-situ collected datasets in underwater environments. We have presented a complete pipeline for underwater object detection, tracking, and counting. We tested the application at different velocities to get the system's minimum and maximum operating velocities. The object detection model is used rather than only color-based object detection. This project's performance has been evaluated utilizing in situ underwater datasets. Vision-based project performance depends on experimental settings (depth, light conditions, backdrop, sensor guidance, etc.) and target types. The way we implemented the system achieves satisfactory results in underwater environments.

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ANNEXURE II

PROJECT ACHIEVEMENTS



Smart India Hackathon 2025 - Team Leader Login Credentials

1 message

SIH <aicte.admin@aicte-india.org>
To: amruthavarshini.2302013@srec.ac.in

Sat, Sep 27, 2025 at 12:22 PM

Smart India Hackathon 2025

Dear AMRUTHA VARSHINI N,

Congratulations, Your team has been nominated as one of the participating teams for Smart India Hackathon 2025.

Your institute has nominated you, as a Team Leader for SUPER NOVA6.

Please note that the idea will be submitted by Team Leader only. The other team members will not receive login credentials.

Please use the following credentials to login into your SIH dashboard and submit your idea.

Team ID	78881
Email ID / Username	amruthavarshini.2302013@srec.ac.in
Password	Lead@7823
User Role	Team Leader
Login URL	https://www.sih.gov.in/signin



TECHgium | Abstract Submission

1 message

<techgium@lts.com>

Wed, Oct 15, 2025 at 11:55 PM

To: amruthavarshini.2302013@srec.ac.in

Abstract Submission Received

Dear amruthavarshini.2302013@srec.ac.in,

We have successfully received your abstract submission for **TECHgium**.



Title: GenAI-Based Automated Electrical Wiring Diagram Generation



PID: TG0913378

Our team will review your submission and notify you of the next steps.
Thank you for your participation!

Best regards,
TECHgium Team

ANNEXURE III

PLAGIARIM REPORT